

Student Dropout Prediction

Applied ML & Intelligent Decision Systems Suite (2025) · Notebook 3 of 4

Yuvan Prasath B · github.com/yuvan-prasath

0.9279

ROC-AUC
Best model score

0.8167

F1 Score
Dropout class F1

4,424

Records
Student records

~32%

Dropout rate
Class imbalance

1. Project Overview

Student dropout is one of the most pressing challenges in higher education, carrying significant consequences for students, institutions, and broader socioeconomic outcomes. Early identification of at-risk students allows academic advisors and institutions to intervene before dropout becomes inevitable — through targeted support, financial assistance, or curriculum adjustments.

This notebook builds a binary classifier to predict whether a student will drop out, using the UCI Predict Students' Dropout and Academic Success dataset. The dataset covers 4,424 students enrolled across multiple undergraduate degrees at a Portuguese higher education institution, with 36 features spanning demographics, socioeconomic background, prior academic qualifications, and first and second semester academic performance. The original dataset contains three outcome classes — Dropout, Enrolled, and Graduate — which were collapsed into a binary target (Dropout vs Non-dropout) to focus on the most actionable prediction problem for early intervention.

2. Methodology

2.1 Data Preprocessing

This dataset is the cleanest of the four notebooks in this suite — no missing values, no string-encoded categorical features requiring label encoding, and well-structured numeric columns throughout. The preprocessing steps were:

- The three-class target (Dropout, Enrolled, Graduate) was binarised: Dropout was assigned label 1 (the at-risk positive class); Graduate and Enrolled were assigned label 0 (non-dropout). This decision prioritises the operationally relevant sub-problem of at-risk detection.
- All 36 features were standardised using `StandardScaler`, as features span significantly different numerical ranges — from age (17-70) to semester grades (0-20) to tuition payment status (0 or 1). Standardisation ensures Logistic Regression converges correctly and prevents large-range features from dominating distance-based computations.
- The scaler was fitted exclusively on the training set and applied to both train and test splits, preventing data leakage.
- SMOTE was applied to the training set only after the stratified 80/20 train-test split, balancing the dropout and non-dropout classes for model training while preserving the real-world distribution in the test set.
- Both the trained model and the fitted scaler were saved as `.pkl` artifacts, since any future inference requires the same scaling transformation to be applied first.

2.2 Models Trained

Three classifiers were trained on the SMOTE-resampled, scaled training data and evaluated on the original imbalanced test set using both ROC-AUC and F1-score:

Model	ROC-AUC	F1 Score	Notes
Logistic Regression	~0.89	~0.76	Linear baseline
XGBoost	0.9279	0.8167	Best model — selected
Random Forest	~0.92	~0.81	Close second

XGBoost achieved the best F1-score of 0.8167 and ROC-AUC of 0.9279, making it the selected production model. F1-score was chosen as the primary ranking metric for this domain: in education intervention settings, both false negatives (missing a student who will drop out) and false positives (flagging a student who would have been fine) carry real costs, and F1 balances both.

3. Key Findings

3.1 Strong Predictive Performance

A ROC-AUC of 0.9279 and F1 of 0.8167 represent strong results for a student dropout prediction task on tabular data. Unlike the healthcare readmission problem in Notebook 1 — where many decisive factors (post-discharge behaviour, home environment) are unobserved — student dropout is well-captured by structured academic records. First and second semester performance data, in particular, carries rich signal about student engagement and academic trajectory, explaining why this dataset yields substantially higher model performance.

3.2 Feature Insights via SHAP

SHAP analysis on the XGBoost model consistently identified academic performance features as the dominant predictors of dropout. Curricular units approved in the 1st semester and 2nd semester were the strongest positive signals for graduation — students who pass fewer units in their first semester are significantly more likely to drop out. Tuition fees up to date (whether a student has paid fees on the expected schedule) was a strong financial stress indicator. Age at enrollment showed moderate importance, with older non-traditional students exhibiting slightly higher dropout risk. Scholarship holder status appeared as a protective factor, with scholarship holders dropping out at lower rates — consistent with the EDA findings.

3.3 Binary Framing as a Design Decision

Collapsing the three-class target into binary was a deliberate modelling decision, not a simplification. Predicting between Dropout, Enrolled, and Graduate is a harder multi-class problem with less actionable output — an institution cannot meaningfully intervene differently for 'might graduate' vs 'currently enrolled'. The binary framing directly answers the question that matters: is this student at risk of not completing their degree? This design choice also enables straightforward threshold tuning: by lowering the classification threshold, advisors can cast a wider net and prioritise recall over precision when early-semester interventions are low-cost.

4. Technical Stack

Category	Tools / Libraries
Language	Python 3.10+
ML Framework	Scikit-learn, XGBoost

Imbalance Handling	imbalanced-learn (SMOTE)
Feature Scaling	StandardScaler (fitted on train, applied to both splits)
Explainability	SHAP (TreeExplainer)
Evaluation	ROC-AUC, F1-score, Classification Report, Confusion Matrix
Visualisation	Matplotlib, Seaborn
Model Persistence	joblib (.pkl for model + scaler)
Dataset	UCI Predict Students Dropout & Academic Success (4,424 records, 36 features)
